Speech Processing

1. A speech signal $s(n)$ is known to have been corrupted by additive noise $\nu(n)$. The noisy speech signal can be written in the time domain as $y(n) = s(n) + \nu(n)$ and in the frequency domain as $Y(l,k) = S(l,k) + V(l,k)$ for time-frame index l and frequency index k.

   a) i) Given a conceptual explanation of what is meant by the spectral variance of a signal in this context. [3]

   ii) Give the mathematical definition for the spectral variance of $y(n)$, denoted $\phi_y(l,k)$. [3]

Solution:
(i)
For any signal $x(n)$ with time-frequency domain representation $X(l,k)$, the spectral variance is a measure of the signal power at time-frame l and frequency k.
(ii)

$$\phi_y(l,k) = E\left[|Y(l,k)|^2\right]$$

   b) In a particular speech enhancement application, it is desired to estimate the noise spectral variance. Now, assume that it is only possible to observe $y(n)$.

   i) One method to compute an estimate of the spectral variance of the noise, $\hat{\phi}_v(l,k)$, employs a Voice Activity Detector (VAD). Describe and explain this technique in detail. [2]

   ii) State one other method to compute an estimate of the spectral variance of the noise. Describe and explain this technique in detail and give mathematical expressions for the estimators. [2]

   iii) Compare and contrast the advantages and disadvantages of these alternative methods. [3]

Solution:
(i)
Voice activity detection (VAD): determine the segments of the signal during which only noise is present. Then update the noise model according to:

$$\hat{\phi}_v(l,k) = \begin{cases} \hat{\phi}_v(l-1,k), & \text{if speech is active} \\ \alpha\hat{\phi}_v(l-1,k) + (1-\alpha)|Y(l,k)|^2, & \text{if speech is not active.} \end{cases}$$

The term α has to be chosen according to the desired level of smoothing and can be time and frequency-varying.
(ii)
Minimum statistics (MS) approach: This technique is based on the assumption that during a speech pause, or within brief periods between words and even syllables, the speech energy is close to zero. As a result, a short-term power spectrum estimate of the noisy signal, even during speech activity, decays frequently due to the noise power. Thus, by tracking the temporal spectral minimum without distinguishing between speech presence and speech absence, the noise power in a specific frequency band can be estimated.

$$\hat{\phi}_y(l,k) = \alpha\hat{\phi}_y(l-1,k) + (1-\alpha)|Y(l,k)|^2$$
$$\hat{\phi}_v(l,k) = \min\left\{\hat{\phi}_y(l,k),\ \hat{\phi}_y(l-1,k),\ \ldots,\ \hat{\phi}_y(l-D+1,k)\right\}$$

(iii)
The VAD method is able to estimate the noise simply and accurately, but only if the VAD is itself accurate and reliable - this is rarely the case, especially in the presence of noise. Unlike the Minimum statistics approach, the VAD method cannot update its noise estimate during speech active segments and therefore cannot normally track time-varying noise as well as the MS method. The MS method is more computationally complex.

c) Consider a device which performs noise reduction on a noisy speech signal. The device uses the method of amplitude spectral subtraction.

i) Show that the output signal of the device at time-frame l and frequency index k can be written

$$Z(l,k) = H(l,k)Y(l,k)$$

and write down an expression for $H(l,k)$ in terms of $\hat{\phi}_v(l,k)$. [3]

ii) It is not known what method the device uses to form the estimate $\hat{\phi}_v(l,k)$. Design a test signal to probe the device such that, by observing the output of the device in response to your test signal input, the method of estimation of $\hat{\phi}_v(l,k)$ can be deduced. [4]

Solution:
(i)
For amplitude spectral subtraction, we have

$$\begin{aligned} Z(l,k) &= Y(l,k) - \sqrt{\hat{\phi}_v(l,k)} \\ &= Y(l,k)\left(1 - \frac{\sqrt{\hat{\phi}_v(l,k)}}{Y(l,k)}\right) \\ &= Y(l,k)H(l,k) \end{aligned}$$

(ii)
Here we are aiming to determine whether or not the noise model is updated continuously or only during non-speech periods. We can consider a test signal containing two speech utterances separated in time by a short pause. To this, we can add the 'noise' of a swept sine wave. If the noise model is updated only during speech pauses, the sine wave will not be significantly suppressed during the speech utterances but will be strongly suppressed during and shortly after the pause. If the noise model is updated continuously then the sine wave will be suppressed throughout. The former case indicates VAD-based noise modelling whereas the later indicates a minimum statistics approach.
[Full marks will be awarded for well-reasoned responses].

2. a) Describe and explain 3 measures for the objective (sometimes called 'instrumental') assessment of the quality of speech in a speech coding application. In your answer, include any relevant mathematical definitions as well as an explanation of each measure. [6]

Solution:
Examples are given below.

- Segmental SNR: computed as the average of $10\log(E_{\text{speech}}/E_{\text{error}})$ where E_{error} is the coding error. It should be mentioned that this measure in not well correlated with subjective opinion and is not good for coding schemes that introduce delays as even a one-sample time shift causes a large change in SNR.

- Itakura Distance: computed as

$$\log\left(\left\langle\frac{P(\omega)}{Q(\omega)}\right\rangle\right)-\left\langle\log\left(\frac{P(\omega)}{Q(\omega)}\right)\right\rangle$$

  where <> indicates the average over frequency.

- Itakura-Saito Distance: computed as

$$\left\langle\frac{P(\omega)}{Q(\omega)}-\log\left(\frac{P(\omega)}{Q(\omega)}\right)-1\right\rangle$$

  Both the Itakura and Itakura-Saito Distances are evaluated over 20 ms frames and averaged. Both may also be calculated directly from the LPC coefficients of the original and coded speech. Neither is a true distance metric since they are asymmetrical.

b) The block diagram of the main elements of a speech coder is shown in Fig. 2.1. Consider a general speech signal $s(n)$ to be encoded.

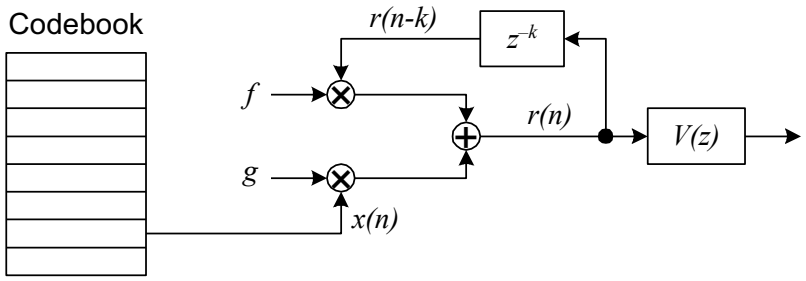


Figure 2.1

i) By making specific reference to Fig. 2.1, identify and explain the operation of the parts of this coder that are responsible for modelling the periodic components and the non-periodic components of the excitation signal $r(n)$.

[2]

Solution:
The periodic component is modelled by the upper feedback loop that involves a delay of z^{-k}. The aperiodic component is modelled by selecting an entry from a fixed codebook, the selection being optimized to give the lowest re-synthesis error.

ii) This coder is designed to determine the values of k and f so as to minimize the energy function

$$E(k,f) = \sum_{n=0}^{N-1} (r(n) - r(n-k)f)^2 .$$

Assume that the value of k is given and $\hat{}$ indicates the optimal value. Show that the optimum values $\hat{f}$ and the corresponding energy function $\hat{E}$ are given by

$$\hat{f}(k) = \frac{\sum_{n=0}^{N-1} r(n)r(n-k)}{\sum_{n=0}^{N-1} r^2(n-k)}$$

and

$$\hat{E}(k) = \sum_{n=0}^{N-1} r^2(n) - \frac{\left(\sum_{n=0}^{N-1} r(n)r(n-k)\right)^2}{\sum_{n=0}^{N-1} r^2(n-k)}.$$

[6]

Solution:
The optimum is found by setting the derivative of E to zero.

$$\frac{\partial E}{\partial f} = -2\sum (r(n) - fr(n-k))\, r(n-k)$$

so that

$$\hat{f} = \frac{\sum r(n)r(n-k)}{\sum r^2(n-k)}.$$

By substituting $\hat{f}$ into E we obtain

$$\begin{aligned}
\hat{E} &= \sum_{n=0}^{N-1} (r(n) - r(n-k)f)^2 \\
&= \sum r^2(n) - 2\hat{f}\sum r(n)r(n-k) + \hat{f}^2 \sum r^2(n-k) \\
&= \sum r^2(n) - 2\frac{(\sum r(n)r(n-k))^2}{\sum r^2(n-k)} + \frac{(\sum r(n)r(n-k))^2}{\sum r^2(n-k)} \\
&= \sum r^2(n) - \frac{(\sum r(n)r(n-k))^2}{\sum r^2(n-k)}.
\end{aligned}$$

iii) In a particular operation of this speech coder, the signal $r(n)$ has the following values and is zero for all other n.

$$\begin{aligned}
r(-100) &= 9 \\
r(-95) &= 4 \\
r(-50) &= 10 \\
r(-45) &= 7 \\
r(0) &= 5 \\
r(5) &= 8
\end{aligned}$$

For $N = 40$, determine $\hat{f}(k)$ and the corresponding values of $\hat{E}(k)$ for $k = 45$ and $k = 50$. [6]

Solution:
In this particular case, there are only 2 terms that contribute to the summations in the numerator and denominator giving:

$$k = 45: \quad \hat{f}(k) = 0.714, \hat{E}(k) = 64$$
$$k = 50: \quad \hat{f}(k) = 0.711, \hat{E}(k) = 13.591.$$

3. a) Describe Linear Predictive Coding (LPC) and its use in speech signal processing applications. Your description should summarise the main concepts in text and, where relevant, diagrams. It is not necessary to give or derive any mathematical formulae. [6]

Solution:
LPC is used as a method of spectral estimation from which characteristic features of speech can be obtained. The LPC coefficients are the denominator coefficients of an all-pole (AR) transfer function. A close link can be derived between the AR system and the vocal tract transfer function. The LPC coefficients may be used in speech coding (often after conversion to Line Spectral Frequencies) and speech recognition (often after conversion to cepstral coefficients). Additional diagrams may be provided for clarification, showing, for example, signal flow through the process, outputs and inputs.

b) A linear predictor for the speech signal $s(n)$ can be written

$$s(n) = Gu'(n) + \sum_{j=1}^{p} a_j s(n-j).$$

Explain what the terms $u'(n)$, G and a_j represent in this expression. [3]

Solution:

$u'(n)$ represents the derivative of glottal flow.
G represents a gain factor.
a_j represent the predictor coefficients, or the coefficients of the denominator polynomial of the vocal tract transfer function.

c) Formulate a time domain expression for the prediction error and hence show that the prediction error is minimum when the following expression is satisfied:

$$\sum_{j-1}^{p} a_j \sum_{n\in\{F\}} s(n-j)s(n-i) = \sum_{n\in\{F\}} s(n)s(n-i) \quad \text{for} \quad i = 1, \ \ldots, \ p$$

where $\{F\}$ denotes the set of speech samples within a frame. [4]

Solution:
Under the assumption that the second term dominates, the prediction error is given by

$$e(n) = s(n) - \sum_{j=1}^{p} a_j s(n-j).$$

The sum squared prediction error is $Q_E = \sum_{n\in\{F\}} e^2(n)$ which is differentiated w.r.t. each a_j giving

$$\frac{\partial Q_E}{\partial a_i} = \sum_{n\in\{F\}} \frac{\partial \left(e^2(n)\right)}{\partial a_i} = -\sum_{n\in\{F\}} 2e(n)s(n-i).$$

The optimum values of a_j are then found from

$$\sum_{n\in\{F\}} e(n)s(n-i) = 0 \quad \text{for} \quad i = 1,\ \ldots,\ p$$

$$\sum_{n\in\{F\}} \left(s(n)s(n-i) - \sum_{j=1}^{p} a_j s(n-j)s(n-i) \right) = 0 \quad \text{for} \quad i = 1,\ \ldots,\ p$$

$$\sum_{j=1}^{p} a_j \sum_{n\in\{F\}} s(n-j)s(n-i) = \sum_{n\in\{F\}} s(n)s(n-i) \quad \text{for} \quad i = 1,\ \ldots,\ p.$$

d) i) Using mathematical notation and conceptual description, define the complex cepstral coefficients of a speech signal and briefly state their important properties for speech processing.

[2]

ii) Derive a recursion to obtain the complex cepstral coefficients from the terms a_j as defined by parts b) and c) of this question.

[4]

iii) Find the complex cepstral coefficients for the case when

$$[a_0,\ a_1,\ a_2,\ a_3,\ a_4] = [1.00,\ -0.20,\ -0.25,\ 0.18,\ 0.40].$$

[1]

Solution:

(i)

Cepstrum is defined as inverse Fourier transform of log spectrum.

$$c_n = \frac{1}{2\pi}\int_{\omega=-\pi}^{\pi} \log\left(V(e^{j\omega})\right) e^{j\omega n} d\omega.$$

It can be computed either from roots of the prediction filter polynomial or from the coefficients of the prediction filter polynomial:

- good at discriminating between different phonemes
- fairly independent of each other
- have approximately Gaussian distributions for a particular phoneme.

(ii)

$$\begin{aligned}
&C(z) = \log(G) - \log(A(z)) \\
&A(z)zC'(z) = -zA'(z) \\
&\left(1 - \sum_{k=1}^{p} a_k z^{-k}\right)\left(z\sum_{m=0}^{\infty} -mc_m z^{-(m+1)}\right) = -z\sum_{n=1}^{p} na_n z^{-(n+1)} \\
&\left(1 - \sum_{k=1}^{p} a_k z^{-k}\right)\left(\sum_{m=1}^{\infty} mc_m z^{-m}\right) = \sum_{n=1}^{p} na_n z^{-n} \\
&\sum_{n=1}^{\infty} nc_n z^{-n} - \sum_{k=1}^{p}\sum_{m=1}^{\infty} mc_m a_k z^{-(m+k)} = \sum_{n=1}^{p} na_n z^{-n} \\
&\sum_{n=1}^{\infty} nc_n z^{-n} = \sum_{n=1}^{p} na_n z^{-n} + \sum_{k=1}^{p}\sum_{n=k+1}^{\infty} (n-k)c_{(n-k)} a_k z^{-n}.
\end{aligned}$$

Now take the coefficient of z^{-n} in the above equation noting that

$$n \geq k+1 \Rightarrow k \leq n-1$$

to give

$$\begin{aligned}
nc_n &= na_n + \sum_{k=1}^{\min(p,n-1)} c_{(n-k)} a_k \\
\Rightarrow c_n &= a_n + \frac{1}{n}\sum_{k=1}^{\min(p,n-1)} c_{(n-k)} a_k.
\end{aligned}$$

(iii)

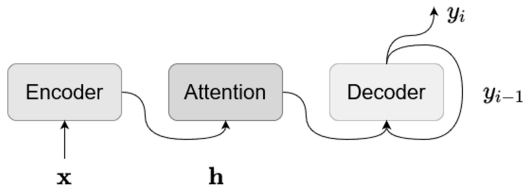
$$c = \begin{pmatrix} 1.0000 & 0.3000 & -0.1167 & -0.0000 & 0.4086 \end{pmatrix}$$

4. a) i) Draw a block diagram showing the main elements of the

Listen-Attend-Spell (LAS)

neural network architecture in a form that could be used in automatic speech recognition. Label the block diagram and explain the meaning of the labels. [3]

Solution:
The block diagram is shown below and must include all 3 elements to receive full marks.



Attention Encoder Decoder model structure
Listen, Attend, Spell

The main elements are (i) the decoder, which takes as input $\mathbf{x}$, the speech input, normally represented as a sequence of feature vectors; (ii) one or more attention head whith attention weights $\mathbf{h}$; and (iii) a decoder which outputs information y_i from which is obtained the word sequence.

ii) The mel frequency scale is used in the process of computing mel-frequency cepstral coefficients (MFCCs). These MFCCs may be used as the input speech features for automatic speech recognition systems. The mel scale is defined as

$$\text{mel}(f) = 2595 \log_{10}\left(1 + \frac{f}{700}\right)$$

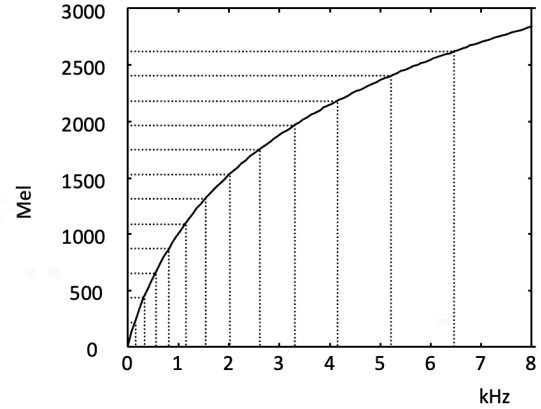
where f is the frequency measured in Hz.

Sketch a graph of the mel scale against frequency for the range $0 \leq f \leq 8000$ Hz. [2]

Explain why the MFCCs are usually a good choice of speech features for automatic speech recognition. [2]

Solution:
The sketch graph is given below. Full numerical labelling on the axis is not required for full marks, but the numerical mapping must be clear from the graph.

The mel scale tracks approximately the frequency resolution of the human perceptual system and is therefore well-suited to extract salient acoustic-linguistic information from the PSD of speech.



b) In a speech recognition system, an observation sequence comprising T frames $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T\}$ is to be compared with a hidden Markov model (HMM) containing S states. For the hidden Markov model, the probability density of generating an observation frame $\mathbf{x}$ from state s is $d_s(\mathbf{x})$ and the transition probability from state i to state j is a_{ij}.

i) Now let $B(t,s)$ be defined as the best (maximum) probability density of generating the sequence $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T\}$ from any sequence of states for which frame 1 is aligned in state 1 and frame t is in state s. Explain how, for $t > 1$, $B(t,s)$ can be expressed in terms of $B(t-1,i)$ for $i = 1,2, \ldots, S$. Indicate the values that should be given to $B(1,i)$. [6]

Solution:
The best path for $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_t\}$ with t in state s must have frame $t-1$ in one of the states, say state i. Since the sub-path $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_t - 1\}$ must also be optimum, we must have:

$$B(t,s) = B(t-1,s) \times a_{is} \times d_s(\mathbf{x}_t)$$

Since $B(t,s)$ represents the probability density of the best path, we must have:

$$B(t,s) = \max_{i=1..S} \left(B(t-1,s) \times a_{is} \times d_s(\mathbf{x}_t) \right)$$

Since we require frame 1 to be in state 1, $B(1,s) = d_1(\mathbf{x}_1)$ if $s = 1$ and 0 otherwise.

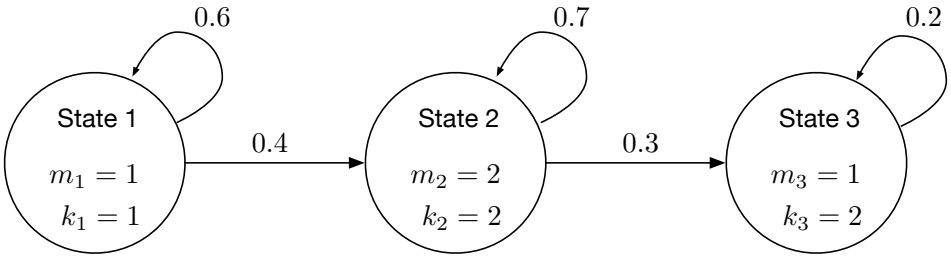


Figure 4.1

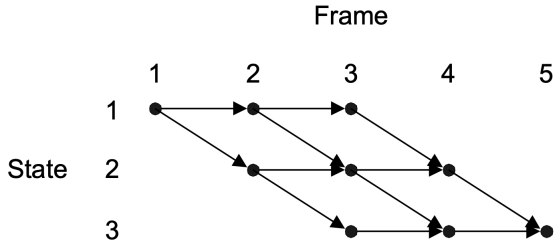
ii) Figure 4.1 shows the transition probabilities for a 3-state hidden Markov model. The input feature vector contains only a single element, and its probability density in frame s is given by:

$$d_s(x) = \frac{\exp\left(-|x - m_s|/k_s\right)}{2k_s}$$

where m_s and k_s are state-dependent parameters given in the figure.

Consider the alignment of the observation sequence $\{x_1, x_2, \ldots, x_5\} = \{1, 1.5, 2, 1, 1\}$ and hence determine the value of $B(5,3)$ and the state sequence to which it corresponds. You should perform all your calculations to 5 decimal places. [7]

Solution:
The possible paths through the State-Frame lattice are:



We have nine lattice points to consider, but five of them have only one possible predecessor.
$B(1,1) = 0.5$
$B(2,1) = 0.5 \times 0.6 \times 0.30327 = 0.09098$
$B(2,2) = 0.5 \times 0.4 \times 0.19470 = 0.03894$
$B(3,1) = 0.09098 \times 0.6 \times 0.18394 = 0.01004$
$B(3,2) = \max\{0.09098 \times 0.4 = 0.03639, 0.03894 \times 0.7 = 0.02726\} \times 0.25 = 0.00910$
$B(3,3) = 0.03894 \times 0.3 \times 0.15163 = 0.00177$
$B(4,2) = \max\{0.01004 \times 0.4 = 0.00402, 0.00910 \times 0.7 = 0.00637\} \times 0.15163 = 0.00097$
$B(4,3) = \max\{0.00910 \times 0.3 = 0.00273, 0.00177 \times 0.2 = 0.00035\} \times 0.25 = 0.00068$
$B(5,3) = \max\{0.00097 \times 0.3 = 0.000290, 0.00068 \times 0.2 = 0.000136\} \times 0.25 = 0.000072$

The best path is 1,1,2,2,3. This may be obtained by tracing the underlined values backwards from $B(5,3)$.